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Supplementary Table 1. Fitbit's timestamped metrics used in this study - resolutions and data processing methods

Data Type / Metric	Temporal Resolution	Processing method / Metric Used
Sleep	Timestamped*: sleep stage classification (light, restless, awake, etc.) during main sleep	1. Sleep Onset (HH:MM) was defined as the first full 5 minutes the user was classified as asleep. 2. Sleep End (HH:MM) was defined as the last time the user was classified as asleep. 3. Sleep Duration (h): The time between Sleep Onset and Sleep End that the user was classified as asleep (light, deep, etc.). 4. Percentage of REM sleep was defined as the sum of all REM classified minutes divided by Sleep Duration 5. Sleep Regularity Index was derived using 30-second binary epochs (sleep/wake) over the entire study time, using formula from Fischer et al[1].
Heart Rate	Timestamped*: ~ every 5 seconds throughout the entire day	Resting Heart Rate equivalent (each night): Median heart rate measurement during all deep sleep stages
Heart Rate Variability (RMSSD, LF/HF Ratio)	Timestamped*: Every 5 minutes during main sleep	Median RMSSD and LF/HF Ratio (each night) during deep sleep stage
Steps	Timestamped*: Up to 1 minute resolution, depends on user activity	Daily Step Count: Sum of all step counts in each date from midnight till midnight
Cardiorespiratory Fitness (VO ₂ Max)	Once per day	If range of values were given, the average was computed
Oxygen Saturation (SpO ₂)	Timestamped*: Every one minute during main sleep	Median during deep sleep stage (every night)
Respiratory rate	Average measurement for each sleep stage in main sleep (light, deep, REM)	Deep sleep stage measurement
Temperature Deviation [°C]	Once per night	

* YYYY-MM-DD HH:MM:SS

Supplementary Table 2. Clinical relevance of wearable data and associated risk stratification rubric for healthy adults

Data Type / Metric	Key Findings	Consideration & Limitations	Risk Category		
			Target (ideal)	Intermediate	High
Sleep					
Sleep Duration (h)	- Exhibits a U-shaped relationship with health outcomes; both short and long durations are associated with higher risk[2], [3], [4], [5]. - Deviations from 7-8 hours are linked to higher risks of mortality, cardiovascular disease, hypertension, depression, anxiety, and other adverse outcomes[2], [3], [4], [5].	- Optimal duration can vary based on age and health status. - These strata apply to habitual nightly sleep.	[7–8]	[5–7] , (8–9]	<5, >9
Standard Deviation of Sleep Duration (h)	High night-to-night variability is an independent risk factor for cardiovascular, metabolic, and psychiatric disorders, regardless of average sleep duration[4], [6], [7].	The optimal measurement window (e.g., weekly vs. monthly standard deviation) is yet to be determined	[0–1]	(1–2]	> 2
Sleep Onset (HH:MM)	Some evidence suggests a broad "optimal" window for sleep onset of approximately 10 PM to 1 AM[4], [7], [8], [9].	Existing studies are limited and do not provide sufficient evidence to define risk thresholds for specific sleep onset times.			
Standard Deviation of Sleep onset, midpoint, and wake-up Time (h)	High variability in sleep timing (onset, midpoint, or wake-up) is consistently linked to adverse cardiovascular, metabolic, and mental health outcomes, regardless of average sleep duration and timing[6], [7], [10], [11].	- The impact of variability may differ between weekdays and weekends (can be captured by using "social jetlag" metric). - Measurement windows in studies vary (days to months).	[0–1]	(1–2]	> 2
REM Sleep Proportion (%)	Current evidence suggests that 20–25% REM sleep is optimal, with low proportion of REM sleep (<15%) associated with increased risks of mortality, cardiovascular disease, and mental health disorders[4], [12], [13].	- The accuracy of consumer wearables for classifying sleep stages has known limitations and varies by device[14], [15], [16], [17]. - Definitive evidence for precise thresholds is yet to be determined.	[20–25 %]	[15–20%)	< 15
Sleep Regularity Index (SRI)	As SRI quantifies sleep-wake consistency within 0-100 range, lower SRI values are strongly associated with increased risks of mortality, cardiovascular disease, metabolic disorders, and mental health conditions, independent of sleep duration and timing[18], [19], [20].	- The SRI is generally not available via vendor APIs and must be calculated from timestamped metrics - Measurement windows in studies vary.	[80–100]	[60–80)	< 60
Heart Rate					
Resting Heart Rate (RHR) [bpm]	Elevated RHR has a robust, dose-dependent association with various cardiovascular disease and mortality, certain cancers, and all-cause mortality[21], [22], [23], [24], [25], [26], [27], [28], [29].	The specific algorithm for calculating RHR can vary by vendor and should be validated.	< 70	[70–75)	≥ 75

		• Values are based on measurements done during daytime. • Strata are generalized and do not account for unique cases such as elite athletes or specific medical conditions.			
Heart Rate Variability (HRV)					
RMSSD / SDNN	Lower HRV (both RMSSD and SDNN) is strongly associated with adverse mental health outcomes such as depression[30] and anxiety[31], lower cancer survival rates[32], [33], increased cardiovascular risk[34] and higher mortality risk[35].	• Definitive thresholds for risk have not been established. • HRV is sex and age dependent; risk can be assessed relative to population norms[36], [37], [38].	consult reference values		
LF/HF Ratio	As LF/HF ratio reflects autonomic nervous system balance, some evidence suggests a higher ratio is linked to stress and adverse mental health outcomes[39], [40], [41], [42], [43].	• The interpretation of the LF/HF ratio is a subject of ongoing scientific debate. • LF/HF is sex and age dependent; risk can be assessed relative to population norms[36], [38].	consult reference values		
Steps					
Daily Step Count	• Strong evidence from meta-analyses[44], [45] and large cohort studies[46], [47] indicates a dose-dependent reduction in all-cause mortality and disease incidence with higher step counts • ~8,000 steps/day appears to be a key threshold for substantial risk reduction[46], [47].	Optimal targets may differ for older adults or clinical populations.	> 8000	[5000–8000]	< 5000
Cardiorespiratory Fitness					
VO ₂ max	• VO₂max is a clinical vital sign and one of the strongest predictors of life expectancy and mortality, across all demographics[48], [49] • Each 1 ml/kg/min increase is associated with a ~9% lower mortality risk[50].	• VO₂max is estimated by wearables, not directly measured; • The accuracy of consumer wearables varies by device[51]; • VO₂max naturally declines with age and differs by sex[48], [52], [53]. • Risk can be assessed relative to population norms[52].	consult reference values		
Oxygen Saturation					
SpO ₂	• Clinically, daytime SpO₂ <95% can indicate cardiopulmonary issues, with <92% showing significant risk[54], [55], while evidence linking nocturnal SpO₂ to health outcomes remains mixed[55].	• Further research is needed to clarify the clinical significance of nocturnal SpO₂ measurements • Investigators should be aware of measurement limitation for different levels of skin pigmentation[56], [57]. • Investigators should examine multiple overnight SpO₂ metrics (e.g., median, minimum, maximum), as	[95–100%]	[92–95%]	< 92%

		well as time series variability analysis[58].			
Respiratory rate					
Respiratory rate	• A core vital sign; an increase of just 3-5 breaths/min can be an early indicator of respiratory distress[59]. • A recent large-scale wearable data study found that nightly respiratory rate ranged from 11.8 to 19.2 breaths per minute in 90% of individuals[60]. • Strata are based on NEWS2 scoring system for clinical deterioration[60]	Wearable measurements are typically an average over the night, or sleep stage, and may not capture acute, transient changes.	[12–20]	(9–12), (20–24)	≤ 9 , ≥ 24
Temperature					
Core Temperature [°C], $ \Delta $	• Body temperature is a key diagnostic indicator; and deviations can signal infection, metabolic dysfunction, or other illnesses. • Risk is non-linear; both low (hypothermia) and high (fever) values are concerning.	• Wearables measure skin temperature, which can differ from core body temperature and be affected by the environmental factors. • Some wearables report deviation from a personal baseline rather than absolute temperature. • We acknowledge other intermediate risk levels are possible but retained the current three for consistency and simplicity.	[36.5–37.5] [0–0.5) and (–0.5, 0)	[35–36.5), (37.5–38.5] [0.5–1) and (–1– –0.5]	< 35 , > 38.5 ≥ 1 or ≤ -1

Square brackets denote inclusive and parentheses exclusive interval endpoints.

Supplementary Table 3. Cluster separation metrics (silhouette coefficients) and variance explained by the first two principal components for sleep and physio-behavioral phenotypes in Cohorts A and B

Cohort	Phenotype Domain	Number of Clusters	Median silhouette [IQR]	PC1 variance (%)	PC2 variance (%)	PC1+PC2 variance (%)
A	Sleep	3	0.277 [-0.1 - 0.5]	53.5	21.1	74.6
A	Physio-behavioral	3	0.214 [-0.02 - 0.45]	42.1	18.1	60.2
B	Sleep	3	0.257 [-0.1 - 0.5]	54.2	23	77.2
B	Physio-behavioral	3	0.315 [0.02 - 0.4]	36.9	25.7	62.6

Supplementary Table 4. Descriptive statistics for the three sleep clusters identified using the Sleep Phenotyping Framework in Cohort A.

Metric	Cluster 1 (n=99) Median [IQR] (% target / % intermediate / % high)	Cluster 1 Mean (SD)	Cluster 2 (n=81) Median [IQR] (% target / % intermediate / % high)	Cluster 2 Mean (SD)	Cluster 3 (n=11) Median [IQR] (% target / % intermediate / % high)	Cluster 3 Mean (SD)
Mean sleep duration (h)	6.92 [6.44–7.46] (40.4 / 59.6 / 0.0)	6.91 (0.69)	7.17 [6.64–7.58] (48.1 / 49.4 / 2.5)	7.12 (0.83)	6.80 [6.37–7.47] (36.4 / 63.6 / 0.0)	6.92 (0.77)
SD sleep duration (h)	1.16 [1.05–1.25] (20.2 / 79.8 / 0.0)	1.14 (0.17)	1.51 [1.40–1.69] (0.0 / 100.0 / 0.0)	1.52 (0.19)	2.21 [2.00–2.49] (0.0 / 27.3 / 72.7)	2.22 (0.40)
SD bedtime onset (h)	1.21 [1.01–1.32] (23.2 / 76.8 / 0.0)	1.19 (0.23)	1.63 [1.43–1.88] (0.0 / 81.5 / 18.5)	1.67 (0.36)	2.59 [2.54–3.00] (0.0 / 9.1 / 90.9)	2.83 (0.59)
REM sleep (%)	21.26 [19.13–23.16] (60.6 / 32.3 / 2.0) ¹	21.03 (2.80)	18.79 [16.32–20.87] (32.1 / 51.9 / 13.6) ²	18.62 (3.55)	15.86 [13.03–18.47] (9.1 / 45.5 / 45.5)	15.77 (3.71)
Sleep Regularity Index	73.03 [70.84–75.82] (6.1 / 93.9 / 0.0)	73.50 (3.97)	64.54 [60.54–67.33] (0.0 / 76.5 / 23.5)	64.02 (4.54)	51.37 [47.42–52.26] (0.0 / 0.0 / 100.0)	48.92 (6.81)

Rounding and precision. Percentages are rounded to one decimal place; they may therefore not sum to exactly 100%.

¹ 5 participants (5.1%) in Cluster 1 had a REM sleep proportion above the target range.

² 2 participants (2.5%) in Cluster 2 had a REM sleep proportion above the target range.

Supplementary Table 5. Descriptive statistics for the three physio-behavioral clusters identified using the Phenotyping Framework in Cohort A.

Metric	Cluster 1 (n=29) Median [IQR]	Cluster 1 Mean (SD)	Cluster 2 (n=28) Median [IQR]	Cluster 2 Mean (SD)	Cluster 3 (n=13) Median [IQR]	Cluster 3 Mean (SD)
Deep LF/HF ratio (age group*)	25.00 [25.00 - 50.00]	41.38 (19.22)	50.00 [50.00 - 75.00]	58.04 (21.57)	25.00 [25.00 - 50.00]	42.31 (23.68)
Deep RMSSD (age group*)	75.00 [50.00 - 100.00]	67.24 (26.81)	25.00 [25.00 - 25.00]	28.57 (8.91)	25.00 [25.00 - 25.00]	26.92 (6.93)
VO ₂ max (mL/kg/min)	40.00 [37.00 - 44.00]	40.79 (5.30)	34.50 [31.00 - 38.00]	34.54 (4.34)	30.00 [28.00 - 33.00]	30.31 (4.03)
VO ₂ max (age group*)	95.00 [95.00 - 95.00]	92.76 (5.28)	90.00 [86.25 - 95.00]	88.39 (8.17)	90.00 [75.00 - 95.00]	81.92 (18.99)
SpO ₂ ≥ 95% (% time)	56.67 [36.79 - 66.25]	54.22 (22.12)	37.40 [17.69 - 70.45]	44.31 (28.30)	64.34 [25.91 - 74.44]	51.73 (26.75)
SpO ₂ 92 to < 95% (% time)	36.42 [26.98 - 47.55]	36.80 (17.03)	37.79 [22.12 - 48.91]	37.61 (17.80)	31.97 [23.33 - 54.61]	39.37 (19.93)
SpO ₂ <92% (% time)	5.58 [3.38 - 13.31]	8.98 (7.81)	13.02 [3.62 - 26.10]	18.08 (17.82)	5.04 [2.68 - 9.09]	8.90 (9.31)
Deep HR <70 bpm (% time)	94.95 [90.75 - 98.33]	93.55 (6.11)	55.07 [41.09 - 75.32]	56.84 (21.16)	7.32 [0.85 - 20.30]	11.69 (12.65)
Deep HR 70 to < 75 bpm (% time)	4.00 [1.38 - 8.22]	5.00 (4.58)	26.85 [18.28 - 32.70]	25.11 (8.61)	29.32 [14.42 - 32.82]	23.58 (13.45)
Deep HR ≥ 75 bpm (% time)	0.75 [0.29 - 1.69]	1.44 (1.81)	12.51 [5.92 - 26.00]	18.05 (15.20)	59.13 [50.38 - 85.58]	64.72 (21.70)
Steps >8000 (% days)	48.20 [25.55 - 65.59]	47.43 (24.92)	43.38 [33.03 - 61.34]	45.17 (20.59)	8.53 [5.90 - 17.88]	11.56 (7.82)
Steps 5000–8000 (% days)	25.34 [20.06 - 32.30]	26.18 (10.87)	27.74 [21.71 - 36.34]	29.05 (10.20)	22.73 [18.09 - 27.82]	24.29 (14.96)
Steps <5000 (% days)	20.67 [9.16 - 38.20]	26.38 (21.30)	19.72 [14.75 - 33.95]	25.78 (19.01)	66.93 [58.08 - 73.38]	64.15 (19.33)

*age group percentile relative to population norms (See Table S2) given as the band's upper bound (e.g. 90 denotes a value between the 75th and 90th percentiles of the reference population).

Supplementary Table 6. Descriptive statistics for the three sleep clusters identified using the Sleep Phenotyping Framework in Cohort B.

Metric	Cluster 1 (n=72) Median [IQR] (% target / % intermediate / % high)	Cluster 1 Mean (SD)	Cluster 2 (n=83) Median [IQR] (% target / % intermediate / % high)	Cluster 2 Mean (SD)	Cluster 3 (n=20) Median [IQR] (% target / % intermediate / % high)	Cluster 3 Mean (SD)
Mean sleep duration (h)	7.42 [7.06–7.81] (61.1 / 38.9 / 0.0)	7.44 (0.55)	6.63 [6.30–7.12] (31.3 / 67.5 / 1.2)	6.65 (0.72)	7.46 [7.00–7.96] (60.0 / 35.0 / 5.0)	7.50 (1.03)
SD sleep duration (h)	0.88 [0.79–1.03] (70.8 / 29.2 / 0.0)	0.91 (0.18)	1.31 [1.17–1.49] (2.4 / 97.6 / 0.0)	1.33 (0.21)	2.17 [1.83–2.47] (0.0 / 35.0 / 65.0)	2.19 (0.46)
SD bedtime onset (h)	0.78 [0.66–0.98] (76.4 / 23.6 / 0.0)	0.82 (0.20)	1.38 [1.14–1.55] (15.7 / 81.9 / 2.4)	1.35 (0.31)	2.12 [1.99–2.81] (0.0 / 25.0 / 75.0)	2.45 (0.82)
REM sleep (%)	20.80 [18.85–23.00] (54.2 / 31.9 / 4.2) ¹	20.87 (3.29)	18.74 [15.96–21.37] (36.1 / 41.0 / 19.3) ²	18.35 (4.52)	18.57 [14.72–20.19] (30.0 / 40.0 / 30.0)	16.90 (5.38)
Sleep Regularity Index	79.52 [76.68–82.03] (47.2 / 52.8 / 0.0)	79.12 (4.15)	71.89 [67.53–74.01] (2.4 / 92.8 / 4.8)	70.80 (5.29)	57.12 [51.43–59.97] (0.0 / 25.0 / 75.0)	54.83 (7.64)

Rounding and precision. Percentages are rounded to one decimal place; they may therefore not sum to exactly 100%.

¹ 7 participants (9.7%) in Cluster 1 had a REM sleep proportion above the target range.

² 3 participants (3.6%) in Cluster 2 had a REM sleep proportion above the target range.

Supplementary Table 7. Descriptive statistics for the three physio-behavioral clusters identified using the Phenotyping Framework in Cohort B.

Metric	Cluster 1 (n=17) Median [IQR]	Cluster 1 Mean (SD)	Cluster 2 (n=12) Median [IQR]	Cluster 2 Mean (SD)	Cluster 3 (n=11) Median [IQR]	Cluster 3 Mean (SD)
Deep LF/HF ratio (age group*)	25.00 [25.00 - 50.00]	35.29 (12.68)	25.00 [25.00 - 50.00]	39.58 (19.82)	25.00 [25.00 - 37.50]	36.36 (20.50)
Deep RMSSD (age group*)	25.00 [25.00 - 50.00]	39.71 (17.81)	37.50 [25.00 - 50.00]	41.67 (19.46)	25.00 [25.00 - 37.50]	36.36 (20.50)
VO ₂ max (mL/kg/min)	37.00 [35.00 - 40.00]	37.29 (5.72)	39.00 [34.75 - 40.25]	38.58 (4.70)	29.00 [27.50 - 30.50]	29.82 (4.31)
VO ₂ max (age group*)	95.00 [90.00 - 95.00]	90.00 (8.66)	95.00 [95.00 - 95.00]	94.17 (1.95)	90.00 [75.00 - 90.00]	79.55 (16.50)
SpO ₂ ≥ 95% (% time)	53.03 [28.44 - 67.16]	49.17 (23.36)	62.35 [31.82 - 66.10]	51.36 (26.42)	30.30 [21.19 - 42.73]	34.09 (21.22)
SpO ₂ 92 to < 95% (% time)	37.93 [26.69 - 40.43]	36.42 (15.88)	29.42 [22.42 - 37.97]	31.69 (16.69)	41.41 [28.80 - 49.54]	39.65 (17.73)
SpO ₂ <92% (% time)	9.34 [6.63 - 18.10]	14.42 (11.83)	11.82 [6.91 - 16.47]	16.95 (18.03)	20.88 [14.11 - 26.04]	26.26 (23.12)
Deep HR <70 bpm (% time)	91.56 [52.02 - 96.93]	76.08 (25.01)	79.75 [72.99 - 88.19]	79.25 (13.93)	6.84 [1.61 - 23.88]	14.10 (15.65)
Deep HR 70 to <75 bpm (% time)	7.64 [2.23 - 26.61]	15.37 (15.33)	15.71 [9.78 - 20.10]	14.57 (8.06)	22.33 [13.73 - 33.39]	23.54 (15.47)
Deep HR ≥ 75 bpm (% time)	1.90 [0.61 - 15.12]	8.56 (10.05)	3.46 [1.01 - 7.44]	6.18 (7.26)	70.23 [39.96 - 84.66]	62.36 (29.83)
Steps >8000 (% days)	74.58 [56.60 - 83.88]	71.53 (18.91)	5.45 [1.29 - 16.63]	9.82 (10.35)	20.33 [14.51 - 26.32]	22.90 (12.23)
Steps 5000–8000 (% days)	20.81 [11.26 - 29.77]	20.85 (14.33)	12.51 [5.85 - 37.08]	21.10 (18.35)	34.09 [25.12 - 40.19]	32.48 (11.89)
Steps <5000 (% days)	4.24 [2.11 - 12.15]	7.62 (7.34)	83.25 [42.07 - 92.83]	69.08 (27.27)	47.96 [37.43 - 54.42]	44.62 (16.05)

*age group percentile relative to population norms (See Table S2) given as the band's upper bound (e.g. 90 denotes a value between the 75th and 90th percentiles of the reference population).

Supplementary Table 8. Within-cohort stability of the k = 3 partition.

Dataset	Bootstrap ARI, mean (95% CI)	Jaccard C1 / C2 / C3	Initialization ARI, mean
Cohort A (Sleep)	0.63 (0.20–0.97)	0.83 / 0.68 / 0.58	0.96
Cohort B (Sleep)	0.72 (0.39–0.98)	0.78 / 0.87 / 0.78	0.99
Cohort A (Physio)	0.65 (0.30–1.00)	0.68 / 0.80 / 0.82	0.98
Cohort B (Physio)	0.74 (0.28–1.00)	0.79 / 0.87 / 0.82	0.98

Bootstrap resampling (1,000 resamples) and initialization sensitivity (100 random seeds). ARI, adjusted Rand index; CI, confidence interval; Jaccard indices are per-cluster overlap of bootstrap co-membership.

Supplementary Table 9. Cross-cohort reproducibility of the phenotype axes and partition.**Panel A. Domain level**

Domain	Tucker ϕ (PC1 / PC2)	Transfer bal. acc. Cohort A to B	Transfer bal. acc. Cohort B to A
Sleep	0.997 / 0.987	0.74	0.80
Physio-behavioral	0.89 / 0.43	0.65	0.66

Tucker congruence coefficient^[61] (ϕ) between cohort-specific PCA loadings; $\phi > 0.95$ indicates equal factors, 0.85–0.95 fair similarity. Label-transfer balanced accuracy: a classifier trained on one cohort applied to the other (chance = 0.33 for three balanced classes).

Panel B. Per phenotype

Domain	Phenotype	Centroid profile r (A vs B)
Sleep	Low-risk	0.92
	Intermediate	0.55
	High-risk	0.97
Physio-behavioral	Low-risk	0.89
	Intermediate	0.71
	High-risk	0.90

Centroid profile r : Pearson correlation between the two cohorts' cluster centroids for the matched phenotype, computed across the shared feature set

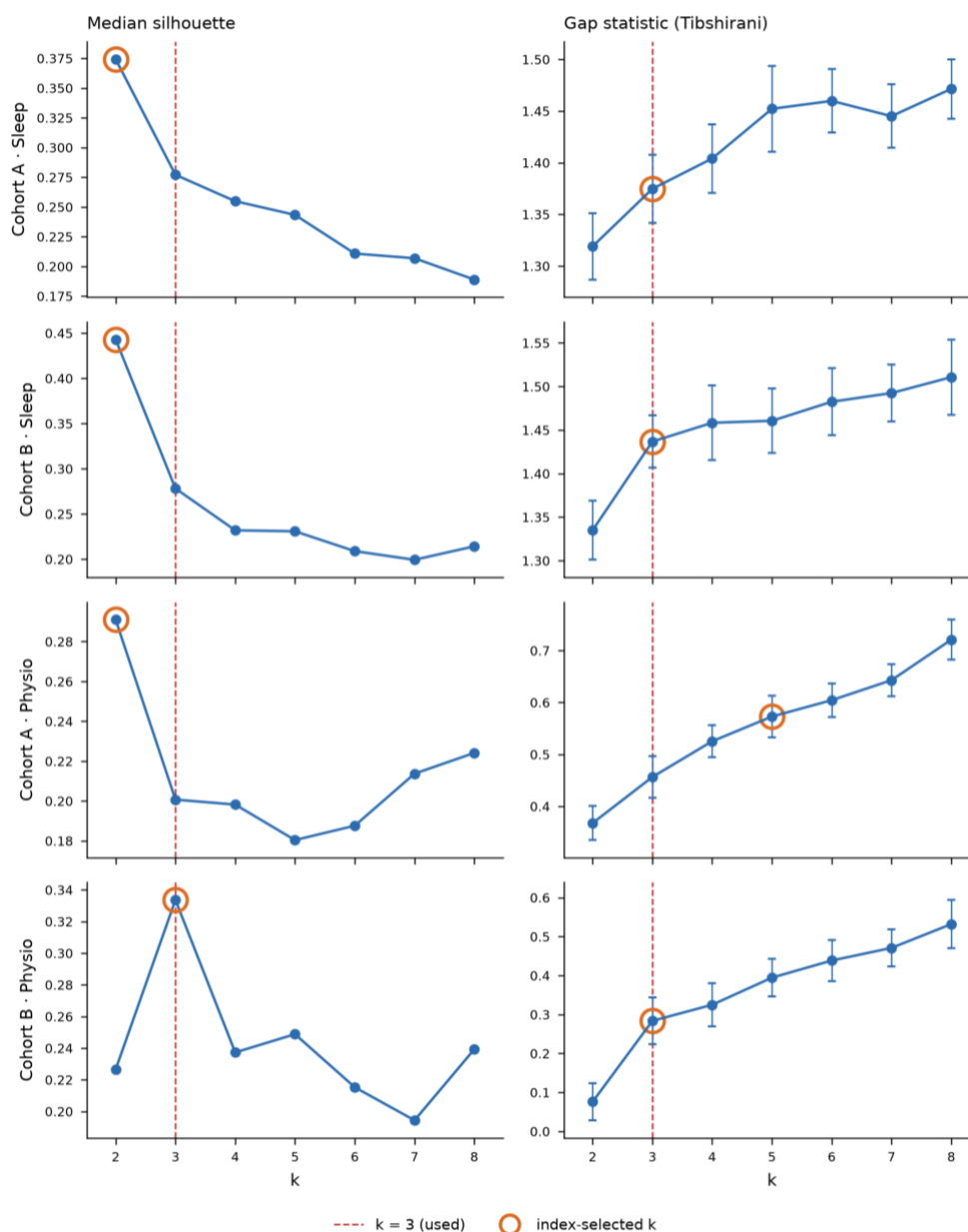
Supplementary Table 10. Sensitivity of the sleep phenotypes to observation-window length.

Quantity	First 90 nights	First 180 nights	Full record (as reported)
Cross-cohort congruence of the phenotype axes			
Tucker ϕ , PC1 (Cohort A vs B)	0.997	0.998	0.997
Tucker ϕ , PC2 (Cohort A vs B)	0.946	0.982	0.987
Variance explained			
PC1, Cohort A	52.2%	52.7%	53.5%
PC1, Cohort B	52.9%	54.0%	54.2%
PC2, Cohort A	20.4%	21.2%	21.1%
PC2, Cohort B	21.7%	22.7%	23.0%
Cluster separation			
Median silhouette, Cohort A	0.271	0.272	0.277
Median silhouette, Cohort B	0.336	0.279	0.257
Agreement of individual assignments with the full-record assignments			
Participants retaining phenotype, Cohort A	87.4%	97.9%	100%
Participants retaining phenotype, Cohort B	65.7%	89.7%	100%
Adjusted Rand index, Cohort A	0.618	0.921	1.000
Adjusted Rand index, Cohort B	0.282	0.695	1.000
Window size			
Median nights used, Cohort A	90	180	250
Median nights used, Cohort B	90	180	330

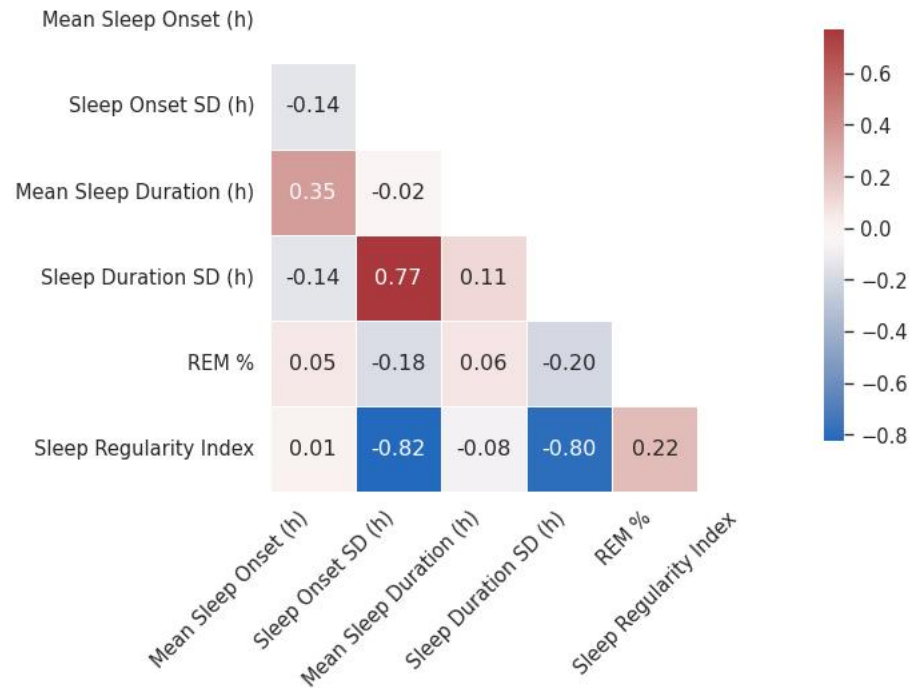
Supplementary Table 11. Age distribution and defining physiological characteristics of the physio-behavioral clusters in both cohorts.

Cohort	Phenotype	n	Age, median (IQR)	Age range	<45	45–59	≥60	Median % of nights with deep-sleep HR ≥ 75 bpm	n in lowest age- and sex-specific normative RMSSD quartile
A	Low-risk	29	55.0 (42–58)	25–77	10	14	5	0.7	5/29
	Intermediate	28	49.5 (44–57)	33–66	8	15	5	12.5	24/28
	High-risk	13	54.0 (51–55)	33–59	3	10	0	59.1	12/13
	All	70	52.5 (42–57)	25–77	21	39	10	6.1	41/70
B	Low-risk	17	56.0 (46–62)	36–70	4	6	7	1.9	9/17
	Intermediate	12	54.5 (52–65)	31–71	2	6	4	3.5	6/12
	High-risk	11	51.0 (46–53)	33–59	2	9	0	70.2	8/11
	All	40	53.0 (47–60)	31–71	8	21	11	11.2	23/40

Supplementary Figure 1. Cluster-number selection across $k = 2-8$ for each cohort and domain (rows), showing the median silhouette (left) and the Tibshirani gap statistic[62] with standard-error bars (right). Red dashed line marks the published $k = 3$; orange rings mark each index's selected k . While $k = 3$ is a clinically motivated granularity aligned with our three-tier risk stratification rather than a claimed global optimum, the gap statistic selects $k = 3$ in both sleep cohorts and in Cohort B physio-behavioral, where silhouette also selects $k = 3$.



Supplementary Figure 2. Pearson correlation matrix of six sleep metrics in Cohort A. The heatmap highlights a strong positive correlation between sleep-onset SD and sleep-duration SD ($r = 0.77$) and strong negative correlations between the Sleep Regularity Index (SRI) and both onset SD ($r = -0.82$) and duration SD ($r = -0.80$). However, SRI incorporates night-to-night concordance of sleep–wake states (including wake-after-sleep-onset events) rather than just the spread of bedtime or duration.



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